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MILITARY SPECIFICATION

CIRCUIT CARD ALARM, LINE SECURITY

PL-1371/FSS-9(V)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an Alarm, Line Security Circuit Card for the Joint-Services Interior Intrusion Detection System (J-SIIDS).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.

MILITARY

MIL-B-131	- Barrier Material, Water Vaporproof, Flexible.
MIL-C-45662	- Calibration Systems Requirements.
MIL-C-52853	- Control Unit, Alarm Set, C-9412/FSS-9(V).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Test: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13221E8554	- Circuit Card Alarm, Line Security, PL-1371/FSS-9(V).

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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3. REQUIREMENTS

3.1 Description. The Alarm Line Security Circuit Card (hereinafter referred to as Alarm Line Security Attachment or ALSA) shall be in accordance with Top Assembly TA13221E8554 and as specified herein. The ALSA shall consist of a circuit card, six complex impedance blocks, and a complex impedance test block. The ALSA is an optional component of the Joint-Services Interior Intrusion Detection System (J-SIIDS). It is designed to mount in the J-SIIDS Control Unit which conforms to MIL-C-52853 and to provide line supervision for the six sensor input lines which enter the control unit.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the ALSA. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more ALSA's as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce ALSA's that comply with the requirements of this specification. Examinations and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified, the Government will conduct any or all of the preproduction examination and tests (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more ALSA's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).

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3.5 Environmental conditions. When tested as specified in 4.6.2, the ALSA shall withstand individual or probable combinations of the environmental conditions specified herein without mechanical or electrical damage or performance degradation.

3.5.1 Operational.

(a) Temperature range of -20° F to 150° F.
(b) Humidity up to 95 percent relative.

3.5.2 Nonoperational and storage.

(a) Temperature range of -30° F to 165° F.
(b) Humidity up to 95 percent relative.
(c) Shock of 20 g's, 11 milliseconds duration and bench handling.
(d) Vibration - transportation conditions.

3.6 Operation. The ALSA provides line supervision to the six sensor inputs to the Control Unit by placing an ac signal on the six incoming sensor line pairs. The phase and amplitude of the return signals, which are established by the terminating complex impedance modules, are continuously monitored by sensing circuits on the circuit card. Changes in the phase and amplitude, of the return signals in excess of a predetermined value causes the ALSA to output a tamper alarm.

3.6.1 ALSA interface. The ALSA circuit card shall interface with the control unit through an 18 pin connector. Interface requirements are as follows:

PIN NO.	FUNCTION
A	NC
B	Sensor F Input
C	NC
D	Sensor E Input
E	NC
F	Sensor D Input
H	NC
J	Sensor C Input
K	Tamper Alarm Output
L	Sensor A Input
M	20 Vdc Input
N	NC
P	NC
R	Sensor B Input
S	Tamper Output Return
T	NC
U	Signal Common (Ground)
V	NC

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3.6.2 Power requirements. The ALSA circuit card shall function as specified herein on regulated voltages of 20 ± 1 volt dc. The current drain shall not exceed 60 ma.

3.6.3 Operating frequency. The operating frequency of the common frequency source to the sensing circuit shall be 300 Hz. The source shall be stable to ± 3 Hz under all operating conditions. The amplitude of the 300 Hz signal placed on each sensor input shall be 10 ± 1 volt peak to peak.

3.6.4 Sensing circuits. Each sensing circuit shall have the capability of being nulled to 400 millivolts or less at its test point when its input is terminated with the complex impedance module.

3.6.5 Tamper alarm criteria. The ALSA shall output a tamper alarm if the absolute value (magnitude) of the impedance vector, associated with the complex terminating impedance of any sensor input, changes in excess of ± 12 percent from the nominal. Changes which exist for 450 milliseconds or greater shall cause a tamper alarm output. Changes which exist for 300 milliseconds or less shall not cause a tamper alarm output.

3.6.6 Output conditions. The output condition of the ALSA circuit card shall correspond to the following impedance levels between pins K and S.

Impedance (ohms)	Output Condition
< 2 K	No Tamper
>100 K	Tamper Alarm

3.6.7 Loss of power output. A tamper alarm output condition shall appear at the ALSA tamper output when power to the ALSA circuit card is interrupted.

3.7 Burn-in. Each ALSA shall be subjected to a 72 hour burn-in to reduce infant mortality failures (see 4.6.3.3).

3.8 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the ALSA shall conform to MIL-STD-461, Test Methods RS03, CS01, CS02, and CS06 and as modified in 3.8.1 and 3.8.2.

3.8.1 Radiated. Radiated susceptibility, Method RS03, shall be conducted in an electric field of 5 V/m or greater for both steady-state and keyed conditions. This field applies to frequencies up to 500 MHz. Above 500 MHz, the electric field shall be 1 V/m.

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3.8.2 Conducted. Conducted interference signals shall be applied to two sets of control unit sensor inputs as follows:

(a) Continuous wave, 30 Hz to 500 KHz with the following exception: EMI tones of 5 V P-P between frequencies of 100 Hz to 800 Hz may cause an alarm.
(b) Interfering spike voltage, 10 spikes 5 to 10 microseconds wide at a repetition rate of 7 to 10 pulses per second on control unit input power leads only. Random phasing of the spikes with respect to the powerline frequency shall be employed.

3.9 Reliability. The specified Mean-Time-Between-Failure (MTBF) of the ALSA shall be 10,000 hours when tested as specified in 4.6.4.

3.10 Identification marking. The ALSA and parts shall be identified in accordance with MIL-STD-130.

3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.6):

Item No.	Description	Identification	Quantity
1	Control Unit	MIL-C-52853(ME)	1

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.12.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan and the basis for acceptance and rejection (see 6.5.4 and 6.9).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction ALSA shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the preproduction ALSA shall be subjected to the tests marked "X" in Column 1 of Table I. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production ALSA will be selected at random by the Government from the ALSA being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2

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of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production ALSA shall not exclude the remaining ALSA's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production ALSA to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production ALSA and shall be cause for refusal by the Government to continue acceptance of production ALSA until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on ALSA previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all ALSA accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level II. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S2.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the major and minor characteristics for possible defects as specified in 4.6.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each ALSA shall be subjected to the tests marked "X" in Column 3 of Table I. Failure of any test shall be cause for rejection.

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4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table I, except that not more than three sample units shall be subjected to the humidity test specified in 4.6.2.6.5. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The ALSA shall be examined as specified herein for the following characteristics:

Number	Characteristic	Req. Paragraph
Major		
101.	The alarm line security circuit card not manufactured in accordance with Top Assembly TA13221E8554.	3.1
102.	Any part or component not in accordance with SQAP requirements as shown on the drawings.	3.1.1
103.	Components missing or not as specified.	3.1.1
104.	Dimensions, that affect form, fit, or function not as specified.	3.1.1
105.	Material not as specified.	3.4
106.	Electromagnetic interference suppression components and network not identical.	3.8
107.	Workmanship not as specified.	3.12
108.	Soldering not in accordance with the requirements of MIL-STD-454.	3.12.1
109.	Identification or special marking missing or illegible.	3.10
Minor		
201.	Dimensions, that do not affect form, fit or function not as specified.	3.1.1
202.	Identification marking not in accordance with MIL-STD-454.	3.11

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, the tests shall be performed under the ambient climatic conditions existing at the place of the test. Unless otherwise specified herein, the ALSA shall operate without maintenance.

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4.6.2.2 Test schedule.

Table I. Test Schedule

Preproduction	Initial Production	Quality Conformance Individual	Quality Conformance Sample	Test	Requirement Para	Test Para
1	2	3	4	5	6	7
X	X	X	X	Functional	3.6	4.6.3.1
X	X	-	X	Low Temperature	3.5	4.6.3.2.2
X	X	-	X	High Temperature	3.5	4.6.3.2.3
X	X	-	-	Shock	3.5	4.6.3.2.4
X	X	-	-	Vibration	3.5	4.6.3.2.5
X	X	-	X	Humidity	3.5	4.6.3.2.6
X	X	X	-	Burn-In	3.7	4.6.3.3
X	X	-	-	Electromagnetic Susceptibility	3.8	4.6.3.4
X	X	-	-	Reliability	3.9	4.6.4

4.6.3 Test methods. Tests shall be conducted in accordance with the Test Method in the Appendix and as specified herein. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.3.1 Functional. The ALSA shall be tested in accordance with the Appendix, Test Method 1. Nonconformance to 3.6 when tested as specified in the Appendix shall constitute failure of this test.

4.6.3.2 Environmental. The ALSA shall be subjected to the following environmental tests specified in MIL-STD-810.

4.6.3.2.1 Functional. Functional testing required during environmental testing shall be conducted as specified in 4.6.3.1. Adjustment of the ALSA sensing circuits as specified in steps g, h, and j of Test Method 1 shall be performed at standard ambient conditions prior to the start of all environmental tests. No adjustment to the ALSA sensing circuits shall be performed during the actual environmental test.

4.6.3.2.2 Low temperature. Low temperature Test Method 502.1, Procedure I, storage temperature -30° F for not less than 48 hours, equipment operating temperature -20° F.

4.6.3.2.3 High temperatures. High temperature Test Method 501.1, Procedure II, except that the nonoperating temperature shall be +165° F. The operating temperature shall be +150° F.

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4.6.3.2.4 Shock.

(a) Basic design: Method 516.2, Procedure I, equipment not operating sawtooth shock pulse, peak value 20 g's, duration 11 milliseconds.
(b) Bench handling: Method 516.2, Procedure V.

Note: Test (a) shall be performed with the ALSA in its shipping container.

4.6.3.2.5 Vibration. Method 514.2, Procedure X, Curve AX, equipment nonoperating.

4.6.3.2.6 Humidity. Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Measurements at the completion of each cycle.
(b) During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle.

The ALSA may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or failure to return the ALSA to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.6.3.2.7 Failure criteria. Nonconformance to 3.5 shall constitute failure of these tests.

4.6.3.3 Burn-in. Each ALSA shall be subjected to a 72 hour burn-in after successfully completing the functional check and prior to performing further tests. During burn-in the ALSA shall be exercised through procedure "k and 1" of the Functional Test nine times per 24 hour period. Burn-in to be performed in the test set described in the Appendix, or a fixture simulating the functions of Procedures "k and 1" of Functional Test, and a 20 Volt power source. Failures (see Procedure "k and 1" of Functional Test) during burn-in shall be analyzed and corrective action taken as necessary. Copies of failure reports shall be forwarded to the contracting officer. Any ALSA failing during burn-in shall be subjected to an additional 72 hours of burn-in.

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4.6.3.4 Electromagnetic susceptibility. The ALSA installed in a control unit shall be subjected to the tests specified in MIL-STD-461, as follows.

4.6.3.4.1 Radiation susceptibility. Test Method RS03. Electric field requirements of RS03 shall be modified and limited as specified in 3.8.1. All leads (sensor, alarm, and power) external to the control unit enclosure shall be exposed to the electric field. The leads shall be not less than 4 feet in length.

4.6.3.4.2 Conducted susceptibility. Test Methods CS01, CS02, and CS06. Interference SPIKE test specified in CS06 shall be modified as specified in 3.8.2.

4.6.3.4.3 Failure criteria. The contractor shall furnish the contracting officer a test plan and a report of the tests specified herein. Disapproval of the test report shall constitute failure of these tests.

4.6.4 Reliability. Using the MTBF specified in 3.9, the ALSA installed in the test set described in the Appendix or a fixture simulating the functions of Procedures "k and 1" of Functional Test and a 20 volt power source, shall be tested as specified herein with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan IV. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.6.4.1 Test procedure. The specified number of ALSA (see 4.4 and 6.2), shall be randomly selected by a Government representative. The testing shall be conducted under ambient laboratory conditions, without temperature, humidity, or vibration cycling. Equipment shall be energized at the nominal primary power input specified, plus 5 or minus 2 percent. Operation shall be continuous throughout the test, with the Functional Test (Procedures "k and 1"), performed a minimum of nine times during each 24 hour period. Reliability tests shall commence after installation and a burn-in period of 72 hours (see 4.6.3.3).

4.6.4.2 Failure criteria. Nonconformance to 3.9 shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select ALSA's at any time during the contract production period and subject these ALSA's to the examination and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on ALSA's selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production ALSA's. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison ALSA to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison ALSA and shall be cause for refusal by the Government to continue acceptance of production ALSA's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on ALSA's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all ALSA's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the characteristics tested in Table II. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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Table II. Examination Schedule-Packaging

Number	Characteristic	Requirement
Major		
110.	Preservation not Level A, Level B, or Commercial as specified (see 6.2).	5.1
111.	Each ALSA Circuit card not preserved for Level A as specified.	5.1.1
112.	Each ALSA Circuit card not preserved for Level B as specified.	5.1.2
113.	Each complete ALSA not preserved for Commercial as specified.	5.1.3
114.	Packing not Level A, Level B, or Commercial as specified.	5.2
115.	ALSA's not packed for Level A as specified.	5.2.1
116.	ALSA's not packed for Level B as specified.	5.2.2
117.	ALSA's not packed for Commercial as specified.	5.2.3
Minor		
203.	Marking for military levels of protection not as specified.	5.3.1
204.	Marking for commercial packaging not as specified.	5.3.2
205.	Special marking not as specified.	5.3.3

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or B or Commercial, as specified (see 6.2).

5.1.1 Level A. Each ALSA Circuit Card shall be wrapped in polyethylene cushioning material conforming to PPP-C-1752, Type II, Class 1, 1/4 inch thick. The cushioning shall be of sufficient size to allow the ends of the material to be stapled or secured with tape to protect the ends of the ALSA Circuit Card from protruding. The wrapped ALSA Circuit Card shall then be placed in a close-fitting heat sealed bag, fabricated from material conforming to MIL-B-131, Class 2. Included air volume shall be kept at a minimum. The loose impedance blocks required for each ALSA shall be placed together in a bag as specified herein. The ALSA Circuit Card and impedance blocks shall then be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the box specification.

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5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A except the bag shall be fabricated from material conforming to L-P-378 with a minimum nominal 4 mils thickness.

5.1.3 Commercial. Each complete ALSA shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A or B, or Commercial, as specified (see 6.2).

5.2.1 Level A. ALSA's, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, Grade B, style optional in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. ALSA's preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c or V3s, style optional in quantities not to exceed the weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. ALSA's preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. Marking shall be military or commercial or special as specified (see 6.2).

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

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6. NOTES

6.1 Intended use. The ALSA is intended for mounting inside a J-SIIDS control unit. It is used as a sensing element protecting the interconnecting lines between the control unit and sensors.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of ALSA's required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required and number of ALSA's to be furnished (see 3.3).
(e) Number of control units if other than that specified to be provided (see 3.11).
(f) Degree of preservation and packing required (see 5.1 and 5.2).
(g) When other than Finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production ALSA's from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish ALSA's conforming to this specification.

6.4 Complex impedance values. Complex impedance values for the complex impedance blocks and for the complex impedance test blocks will be furnished for specific procurement functions on an "as need to know basis" and as directed by the contracting officer.

6.5 Definitions. The following definitions shall apply throughout this specification.

6.5.1 False alarm. An alarm generated by stimuli other than those for which the control unit is designed to alarm.

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6.5.2 Alarm signals.

6.5.2.1 Intrusion alarm. Alarm generated by the sensor subsystem in response to an intrusion or attempted intrusion of the control zone.

6.5.2.2 Tamper alarm. Alarm caused by an attack on the system itself (tamper or line supervisor alarms) and failures which cause an alarm.

6.5.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any component, physical change which impairs the mechanical integrity of the ALSA, evidence of delamination or water penetration into integrated circuits, printed ciruit boards, or external components, or operational requirements resulting in noncompliance of the ALSA with the provisions of this specification.

6.5.4 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.6 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.11, when applicable (see 6.2).

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each ALSA.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of ALSA's.

6.9 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

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6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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Preparing activity:
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APPENDIX

TEST METHODS

FOR

CIRCUIT CARD ALARM, LINE SECURITY

PL-1371/FSS-9(V)

10. Scope. The test methods contained in this Appendix are to determine whether ALSA's procured under this specification conform to the requirements specified. Figure A-1 contains data for fabrication of a test set which is required to facilitate testing of that ALSA.

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TEST METHOD NO. 1

FUNCTIONAL TEST

1. TEST APPARATUS:

   (a) Test Set - See Figure A-1.
   (b) Oscilloscope - Tektronix 422 or equivalent.
   (c) 2 ea Regulated power supplies - HP6202 or equivalent.
   (d) Frequency counter - HP5532A or equivalent.
   (e) Digital Voltmeter - Fairchild 7000 or equivalent.

2. TEST PROCEDURE:

   (a) Install ALSA circuit card under test into connector J1 of test set.
   (b) Install the six complex impedance blocks and the impedance test block under test in test set as shown.
   (c) Set power supplies to +20 volts and +10 volts respectively, turn off, and connect to test set at TB-P.
   (d) Before turning power supplies back on ensure test set switches are in following position:

      (1) Switch 7 should be closed.
      (2) Switch 8, 9, 10 and 11 should be set to position A.

   (e) Connect frequency counter between TP-2 (see Figure A-2) of ALSA card and ground. Observe 300 $\pm$ 3 Hz. Record.
   (f) Disconnect counter and connect oscilloscope to TP-2. Observe 300 $\pm$ Hz, 10 $\pm$ 1 V peak to peak sine wave. Record amplitude. Disconnect oscilloscope from TP-2.
   (g) Adjust sensing circuit for sensor input A as follows:

      (1) Connect oscilloscope to TP-1.
      (2) Adjust potentiometer R3 and R6 alternately until a minimum signal of 400 mv peak to peak or less is obtained at TP-1.

   (h) Adjust sensing circuits for sensor inputs B through F by repeating step (g) using potentiometers and test points on ALSA cards as follows:

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Sensor Input	Test Point	Potentiometers
B	501	R503, R506
C	201	R203, R206
D	301	R303, R306
E	401	R403, R406
F	101	R103, R106

(j) Repeat steps (g) and (h) to ensure all sensing circuits are nulled to 400 mv (peak to peak). Record signal level at each test point.

Note: The LED at this time should be on indicating a non-tamper condition.

(k) Momentarily (at least 1 second) depress switch S1 opening sensor circuit A. The LED should turn off and stay off as long as the circuit is open indicating a tamper alarm condition.

(l) Repeat step (k) for switches 2 through 6 thereby opening circuits B through D. The LED should turn off each time a circuit is opened.

(m) Open switch 7 for 5 seconds. (This simulates a condition normally seen by the Control Unit and which should not affect the ALSA sensing circuits.) The LED shall remain on.

(n) Momentarily place switch 11 to position B and back to position A. This action causes the test impedance block to be switched in parallel with sensor input A. This simulates 12 percent change in the complex impedance normally seen by sensing circuit A for 450 milliseconds. The LED shall turn off indicating tamper alarm condition.

(o) Set switch 10 to position B. Momentarily place switch 11 to position B and back to A. This action repeats action of step (n) but for only 290 msec. The criteria for the tamper alarm condition is not met and the LED shall not turn off.

(p) Set switch 9 to position B. Set switch 10 to position A. Repeat step (n). This simulates a 12 percent change in the complex impedance normally seen by sensing circuit A. The LED shall turn off.

(q) Repeat step (o). The LED shall remain on.

(r) Repeat steps (n) through (q) for sensor inputs B through F by selecting appropriate input with switch 8.

(s) Return switch 8 to position A.

(t) Turn off the 20 volt power supply. This simulates loss of power to ALSA card. The LED shall turn off.

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(u) Turn on power supply. The LED shall turn on.
(v) Turn meter selector switch on 20 volt power supply to current position (ma) or insert current meter into positive 20 volt lead to test set. Repeat step (k). Meter reading shall not exceed 60 ma during any portion of test. Record.
(w) Set 20 volt power supply to 19 volts. Repeat step (v).
(x) Set 20 volt power supply to 21 volts. Repeat step (v).

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NOTES:

1. SELECT R1 AND C1 FOR POSITIVE PULSE OF 450 ± 10MSEC DURATION AT PIN 4 OF ONE SHOT UI-A AND B.
2. SELECT R2 AND C2 FOR POSITIVE PULSE OF 290 ± 10MSEC DURATION AT PIN 11 OF ONE SHOT UI-C AND D.
3. ALL RESISTORS 1/4 WATT, 5% TOLERANCE UNLESS OTHERWISE INDICATED.
4. CAPACITORS 20 VOLT MINIMUM, 10% TOLERANCE.
5. RELAY K1 IS A MODEL # 1A12AHH-X MANUFACTURED BY ELECTRONICS APPLICATIONS CO.

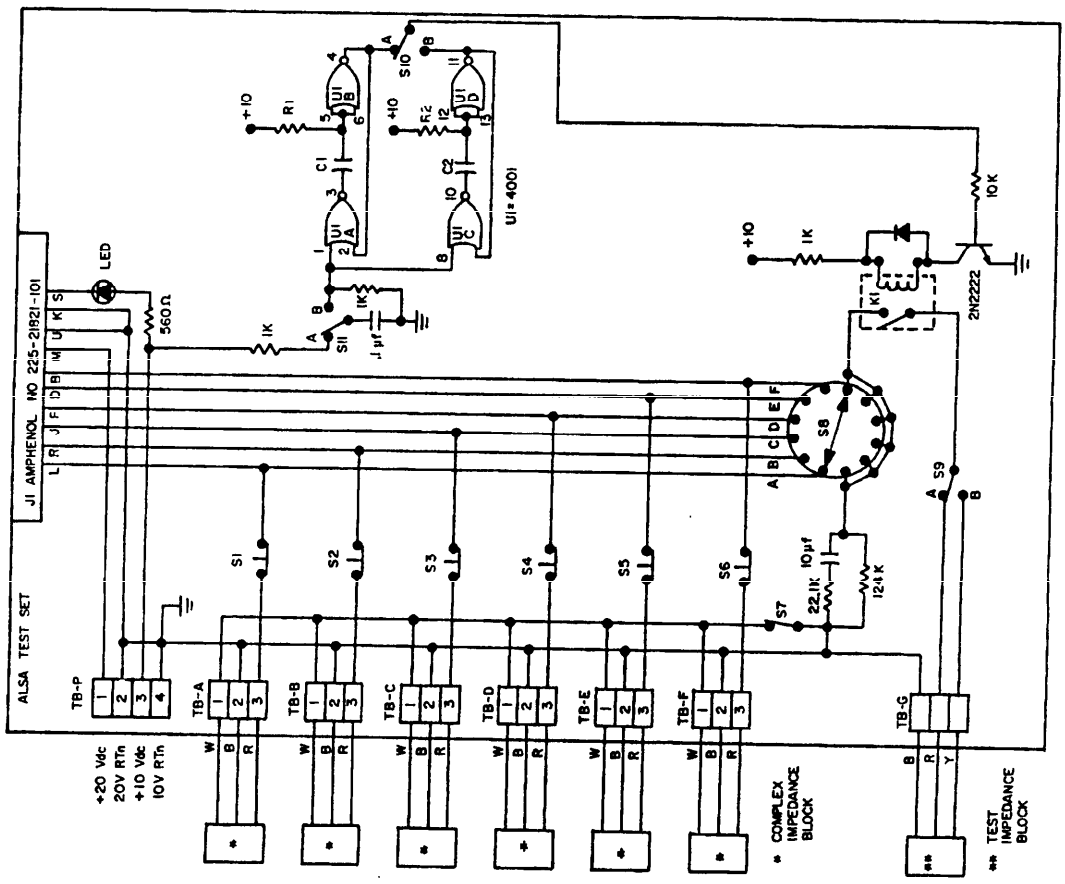
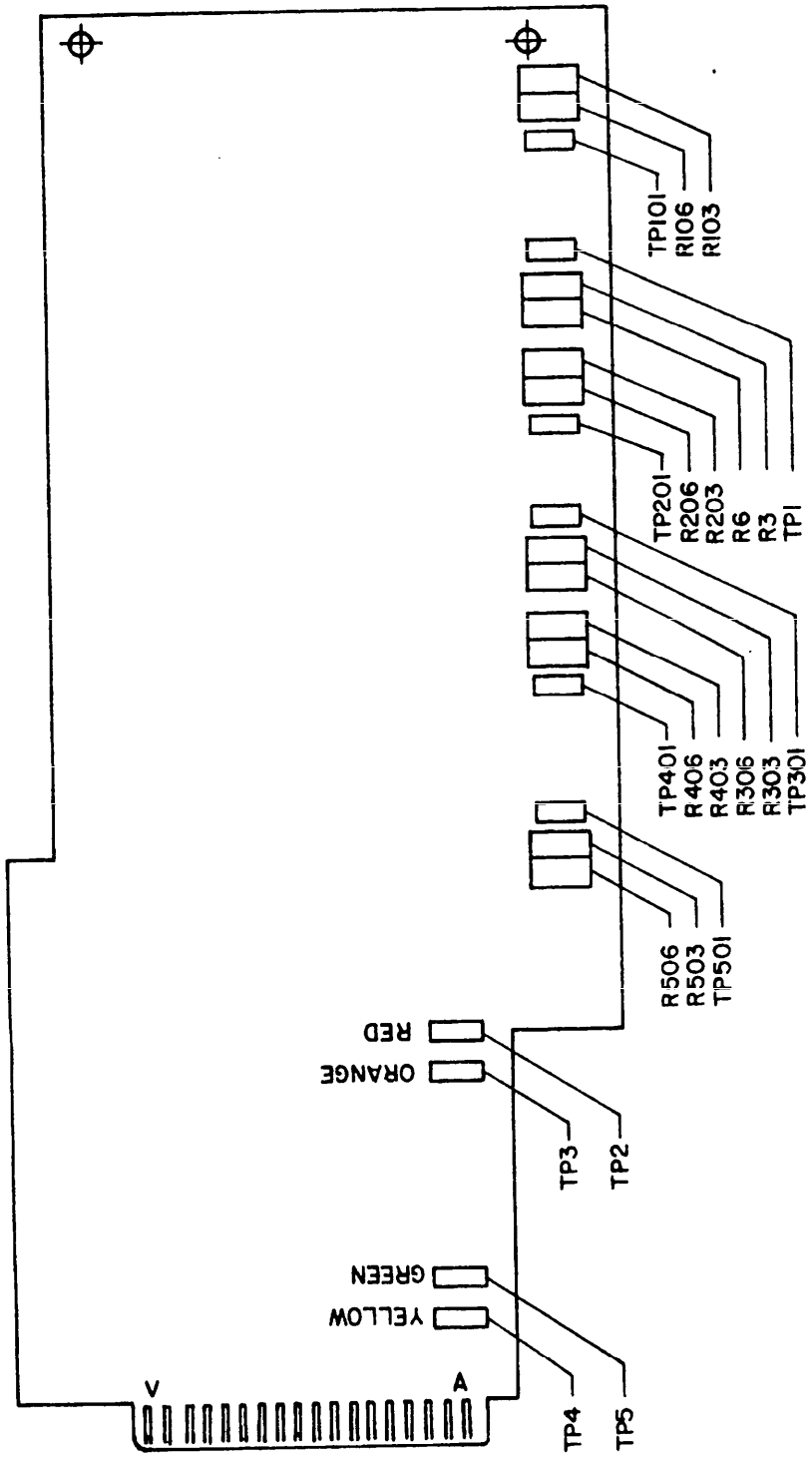


FIGURE A-1. ALSA test set.

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FIGURE A-2. ALSA test points and adjustment locations.

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-52917A(ME) Circuit Card Alarm, Line Security PL-1371/FSS-9(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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